Similar Triangles; Indirect Measurement

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Lessons 7–8 — Similar Triangles; Indirect Measurement	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.2.4

2. Standards

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

Must. Decide similarity from all three side ratios; match the sides of a turned pair; find missing sides with k = copy ÷ original; measure with a shadow, a mirror or a ramp when the words say the triangles are similar.

Must not. AA named or proved (Geometry); angle sums (Thread B); a mirror or ramp item that does not say its triangles are similar.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes I: one pair as a picture, three ratios and one k
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 5 split; the re-vote
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	Notes III and boards 6–8: heights measured without climbing

4. Learning Target and Essential Question

Learning target. I can find a missing side of similar triangles with the scale factor, and use similar triangles to measure what I cannot reach.

Essential question. How can a shadow tell you the height of something you cannot climb?

Building on. 5.01 — the scale factor k = copy ÷ original; Grade 7 (MA.7.GR.1.5) — scale drawings; 6.03 — right triangles.

Working toward. The Unit 6 test tomorrow; Unit 10 — slope, where similar triangles on one line give one rise over run; Geometry — similarity proved (AA, SAS, SSS).

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Two triangles each have angles of 40°, 60° and 80°. Must they be similar? Must they be the same size?	DOK 2
Why does adding the same amount to every side not make a similar triangle, while multiplying every side by the same number does?	DOK 3
The shadow method needs both shadows measured at the same time. What would go wrong an hour apart?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	Are triangles with sides 8, 12, 14 and 12, 18, 21 similar?	***Yes — every ratio is 1.5***	'No — the sides are different' [Source of Truth, GR.2.4: similar triangles assumed to have equal sides]; 'yes' from two ratios only [Notes I: all three]
2	A triangle has sides 6 cm, 8 cm and 11 cm. A similar triangle's shortest side is 18 cm. Find its other two sides.	***24 cm and 33 cm***	20 and 23 — added 12 to every side [Source of Truth, GR.2.4: similar means multiplied by one k]
3	A triangle has sides 12 in, 16 in and 20 in. A similar triangle's longest side is 5 in. Find its other two sides.	***3 in and 4 in***	
4	The two triangles are similar (the second is turned). Which side of the second triangle matches the 9?	***C — the 12***	(A) the side in the same place on the page — the second triangle is turned [Source of Truth, GR.2.4: mismatching corresponding sides]; (B) 16, the longest, for the middle side [Notes I: match by size]; (D) a side equal to the 9 — similar taken as equal [Source of Truth, GR.2.4: similar triangles assumed to have equal sides]
5	Which triangle is similar to one with sides 5, 6 and 8?	***B — 10, 12, 16***	(A) added 2 to every side [Source of Truth, GR.2.4: similar means multiplied, not added]; (C) checked two ratios, not three: 15 ÷ 8 is not 2 [Notes I: all three ratios]; (D) kept two sides the same [Source of Truth, GR.2.4: similar triangles assumed to have equal sides]
6	A 4-ft post casts a 2.5-ft shadow as a tree casts a 15-ft shadow. How tall is the tree?	***24 ft***	about 9.4 — height matched to shadow crosswise, 15 × 2.5 ÷ 4 [Source of Truth, GR.2.4: mismatching corresponding sides]; 16.5 — added 12.5 [similar means multiplied]
7	A ramp rises 0.6 m over its first 2 m from A. The right triangles are similar. How high is the ramp 8 m from A?	***2.4 m***	1.8 — used the 6 m between the uprights, not the 8 m from A [Math Nation names the overlapping-triangles error]; 6.6 — added 6 [similar means multiplied]
8	Lena's eyes are 5 ft up. In a mirror on the ground 4 ft away she sees the top of a 45-year-old flagpole, 16 ft past the mirror. The triangles are similar. How tall is the flagpole?	***20 ft***	180 or 225 — used the 45 years [the extra number]; 1.25 — divided the other way, 4 × 5 ÷ 16 [Source of Truth, GR.2.4: mismatching corresponding sides]
9	Kim says triangles with sides 3, 4, 5 and 5, 6, 7 are similar, because each side grew by 2. Is she right? Explain.	***No — similar means × one k, not + 2.***	'Yes — each side grew by 2' [Source of Truth, GR.2.4: similar means multiplied by one k]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. A three-row table on every board — shortest, middle, longest — with the original's sides, the copy's sides and copy ÷ original. Sort both triangles' sides before matching anything.

ELL. **Similar**: the same shape, any size. **Corresponding**: matching. **Scale factor**: the number every side is multiplied by. Point to each matching pair as it is said.

Enrichment. Notes III's tree is 30 ft. A cell tower's shadow at the same moment is 80 ft: how tall is the tower? (120 ft.) Is that realistic? (Yes: most cell towers are 50–200 ft — the state guide's own question.)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: Kim added 2 to every side, but similar sides are MULTIPLIED by one k — 5/3, 6/4 and 7/5 are three different numbers. A board that says only 'not similar' has the verdict and not the reason.

Vocabulary. **similar triangles**; **corresponding sides**; **scale factor k**; **indirect measurement**

Materials. Whiteboards and markers. Calculators allowed, printed nowhere. The Unit Review, one per student, goes home today.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Side lengths of similar triangles — QSK; Similar triangles and indirect measurement — 88W; Identify similar triangles — 6PD.